



Retail Operations: Data Cleaning & Sales Dashboard

Transforming raw data into actionable insights: Complete data cycle analysis for retail operations

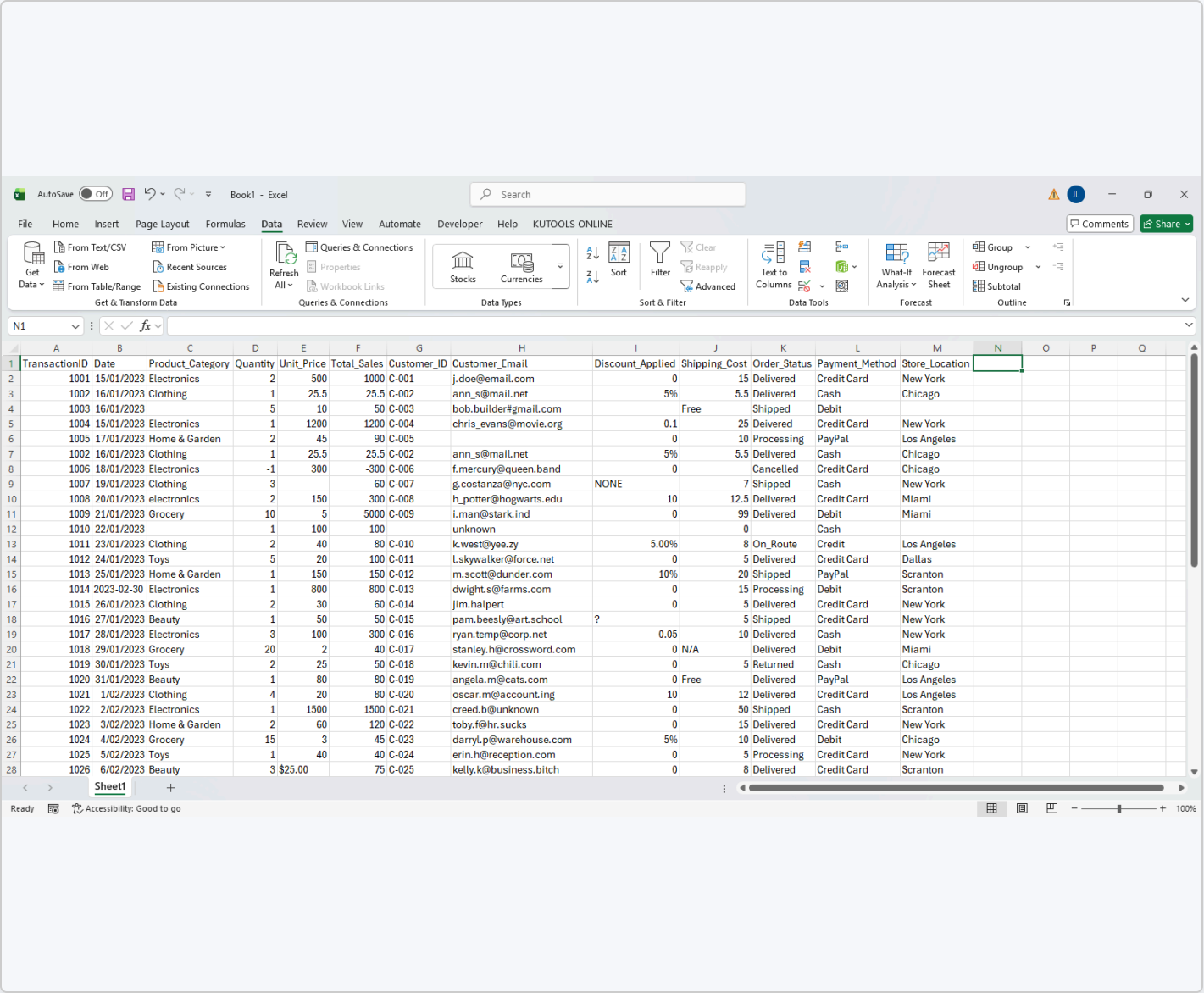
 Excel

 Power BI

 DAX

January - March 2023
Data Analysis Project

The Challenge: Raw Data in Excel



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
	TransactionID	Date	Product_Category	Quantity	Unit_Price	Total_Sales	Customer_ID	Customer_Email	Discount_Applied	Shipping_Cost	Order_Status	Payment_Method	Store_Location				
1	1001	15/01/2023	Electronics	2	500	1000	C-001	j.doe@email.com		0	15	Delivered	Credit Card	New York			
2	1002	16/01/2023	Clothing	1	25.5	25.5	C-002	ann_s@mail.net	5%	5.5	Delivered	Cash	Chicago				
3	1003	16/01/2023		5	10	50	C-003	bob.builder@gmail.com		Free	Shipped	Debit					
4	1004	15/01/2023	Electronics	1	1200	1200	C-004	chris_evans@movie.org	0.1	25	Deivered	Credit Card	New York				
5	1005	17/01/2023	Home & Garden	2	45	90	C-005		0	10	Processing	PayPal	Los Angeles				
6	1002	16/01/2023	Clothing	1	25.5	25.5	C-002	ann_s@mail.net	5%	5.5	Delivered	Cash	Chicago				
7	1006	18/01/2023	Electronics	-1	300	-300	C-006	f.mercury@queen.band	0		Cancelled	Credit Card	Chicago				
8	1007	19/01/2023	Clothing	3		60	C-007	g.costanza@nyc.com	NONE	7	Shipped	Cash	New York				
9	1008	20/01/2023	electronics	2	150	300	C-008	h.potter@hogwarts.edu		12.5	Delivered	Credit Card	Miami				
10	1009	21/01/2023	Grocery	10	5	5000	C-009	i.man@stark.ind	0	99	Delivered	Debit	Miami				
11	1010	22/01/2023		1	100	100		unknown		0		Cash					
12	1011	23/01/2023	Clothing	2	40	80	C-010	k.west@yee.zy	5.00%	8	On_Route	Credit	Los Angeles				
13	1012	24/01/2023	Toys	5	20	100	C-011	l.skywalker@force.net	0	5	Delivered	Credit Card	Dallas				
14	1013	25/01/2023	Home & Garden	1	150	150	C-012	m.scott@dunder.com	10%	20	Shipped	PayPal	Scranton				
15	1014	2023-02-30	Electronics	1	800	800	C-013	dwight.s@farms.com	0	15	Processing	Debit	Scranton				
16	1015	26/01/2023	Clothing	2	30	60	C-014	jim.halpert	0	5	Delivered	Credit Card	New York				
17	1016	27/01/2023	Beauty	1	50	50	C-015	pam.beesly@art.school	?	5	Shipped	Credit Card	New York				
18	1017	28/01/2023	Electronics	3	100	300	C-016	ryan.temp@corp.net	0.05	10	Delivered	Cash	New York				
19	1018	29/01/2023	Grocery	20	2	40	C-017	stanley.h@crossword.com	0	N/A	Delivered	Debit	Miami				
20	1019	30/01/2023	Toys	2	25	50	C-018	kevin.m@chili.com	0	5	Returned	Cash	Chicago				
21	1020	31/01/2023	Beauty	1	80	80	C-019	angela.m@cats.com	0	Free	Delivered	PayPal	Los Angeles				
22	1021	1/02/2023	Clothing	4	20	80	C-020	oscar.m@account.ing	10	12	Delivered	Credit Card	Los Angeles				
23	1022	2/02/2023	Electronics	1	1500	1500	C-021	creed.b@unknown	0	50	Shipped	Cash	Scranton				
24	1023	3/02/2023	Home & Garden	2	60	120	C-022	toby.f@hr.sucks	0	15	Delivered	Credit Card	New York				
25	1024	4/02/2023	Grocery	15	3	45	C-023	darryl.p@warehouse.com	5%	10	Delivered	Debit	Chicago				
26	1025	5/02/2023	Toys	1	40	40	C-024	erin.h@reception.com	0	5	Processing	Credit Card	New York				
27	1026	6/02/2023	Beauty	3	\$25.00	75	C-025	kelly.k@business.bitch	0	8	Delivered	Credit Card	Scranton				

Detected Issues:

Mixed Formats

- Dates stored as text
- Shipping_Cost with values "Free" and "N/A"
- Discount_Applied as "5%" or "NONE"

Inconsistencies

- Order_Status with multiple variants
- Payment_Method without standardization

Missing Values & Errors

- Atypical emails and empty cells
- Totals inconsistent with calculations

Potential Duplicates

- Records with the same Transaction_ID

Disparate Standards

- Store_Location with different conventions

Cleaning Objective: standardize types and values, deduplicate, validate calculations.

The Process: Cleaning & Transformation

1 Column Profiling & Typing

Initial analysis of data types, ranges, and value distribution. Data dictionary creation.

2 Normalization

Date conversion to dd/mm/yyyy format. Currency value transformation. Standardization.

3 Quality Control

Duplicate removal. Handling null values. Validation rules for integrity.

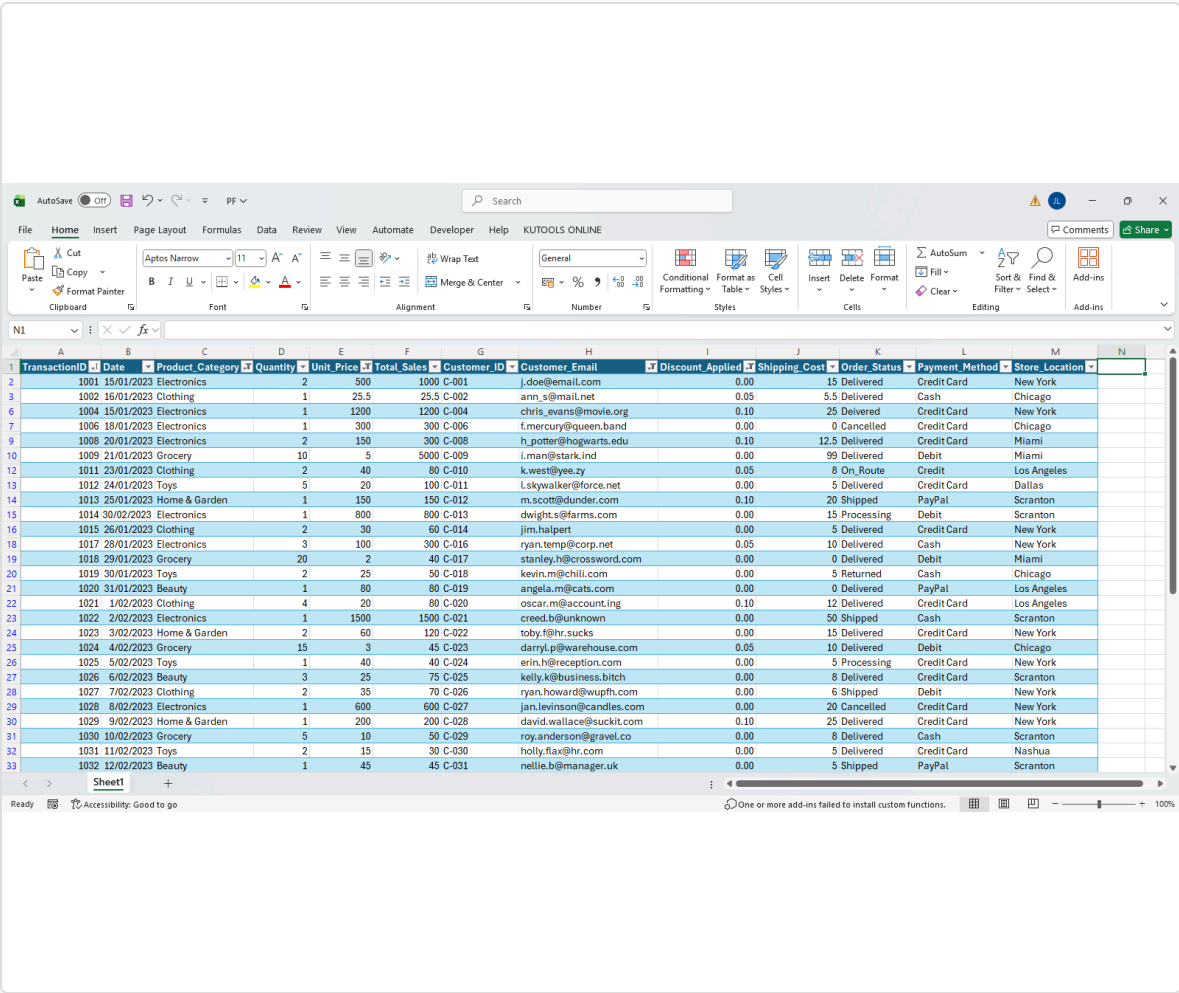
4 Calculations

Total_Sales = Quantity * Unit_Price * (1 - Discount) + Shipping_Cost. Cross-verification.

5 Data Model

Final structuring. Preparation for Power BI and relationship creation for analysis.

Result: Clean & Structured Data



TransactionID	Date	Product Category	Quantity	Unit Price	Total Sales	Customer ID	Customer Email	Discount Applied	Shipping Cost	Order Status	Payment Method	Store Location
1001	15/01/2023	Electronics	2	500	1000	C-001	j.doe@email.com	0.00	15	Delivered	Credit Card	New York
1002	16/01/2023	Clothing	1	25.5	25.5	C-002	ann_s@mail.net	0.05	5.5	Delivered	Cash	Chicago
1004	15/01/2023	Electronics	1	1200	1200	C-004	chris_evans@movie.org	0.10	25	Delivered	Credit Card	New York
1006	18/01/2023	Electronics	1	300	300	C-006	f.mercury@queen.band	0.00	0	Cancelled	Credit Card	Chicago
1008	20/01/2023	Electronics	2	150	300	C-008	h.potter@hogwarts.edu	0.10	12.5	Delivered	Credit Card	Miami
1009	21/01/2023	Grocery	10	5	5000	C-009	i.man@stark.ind	0.00	99	Delivered	Debit	Miami
1011	23/01/2023	Clothing	2	40	80	C-010	k.west@yee.zy	0.05	8	On Route	Credit	Los Angeles
1012	24/01/2023	Toys	5	20	100	C-011	l.skywalker@force.net	0.00	5	Delivered	Credit Card	Dallas
1013	25/01/2023	Home & Garden	1	150	150	C-012	m.scott@dunder.com	0.10	20	Shipped	PayPal	Scranton
1014	30/02/2023	Electronics	1	800	800	C-013	d.wight@farms.com	0.00	15	Processing	Debit	Scranton
1015	26/01/2023	Clothing	2	30	60	C-014	j.m.halpert	0.00	5	Delivered	Credit Card	New York
1017	28/01/2023	Electronics	3	100	300	C-016	ryan.temp@corp.net	0.05	10	Delivered	Cash	New York
1018	29/01/2023	Grocery	20	2	40	C-017	stanley.h@crossword.com	0.00	0	Delivered	Debit	Miami
1019	30/01/2023	Toys	2	25	50	C-018	kevin.m@chili.com	0.00	5	Returned	Cash	Chicago
1020	31/01/2023	Beauty	1	80	80	C-019	angela.m@cats.com	0.00	0	Delivered	PayPal	Los Angeles
1021	1/02/2023	Clothing	4	20	80	C-020	oscar.m@accounting	0.10	12	Delivered	Credit Card	Los Angeles
1022	2/02/2023	Electronics	1	1500	1500	C-021	creed.b@unknown	0.00	50	Shipped	Cash	Scranton
1023	3/02/2023	Home & Garden	2	60	120	C-022	toby.f@hr.sucks	0.00	15	Delivered	Credit Card	New York
1024	4/02/2023	Grocery	15	3	45	C-023	darryl.p@warehouse.com	0.05	10	Delivered	Debit	Chicago
1025	5/02/2023	Toys	1	40	40	C-024	erin.h@reception.com	0.00	5	Processing	Credit Card	New York
1026	6/02/2023	Beauty	3	25	75	C-025	kelly.k@business.bitch	0.00	8	Delivered	Credit Card	Scranton
1027	7/02/2023	Clothing	2	35	70	C-026	ryan.howard@vuphf.com	0.00	6	Shipped	Debit	New York
1028	8/02/2023	Electronics	1	600	600	C-027	jane.lewisson@candles.com	0.00	20	Cancelled	Credit Card	New York
1029	9/02/2023	Home & Garden	1	200	200	C-028	david.wallace@suckit.com	0.10	25	Delivered	Credit Card	New York
1030	10/02/2023	Grocery	5	10	50	C-029	roy.anderson@gravel.co	0.00	8	Delivered	Cash	Scranton
1031	11/02/2023	Toys	2	15	30	C-030	holly.flax@hr.com	0.00	5	Delivered	Credit Card	Nashua
1032	12/02/2023	Beauty	1	45	45	C-031	nellie.b@manager.uk	0.00	5	Shipped	PayPal	Scranton

Techniques:

 Power Query

 Replace Values

 Trim/Clean

 Calculated Columns

Results: Sales & Operations Dashboard

\$ Total Sales

\$16,194.50

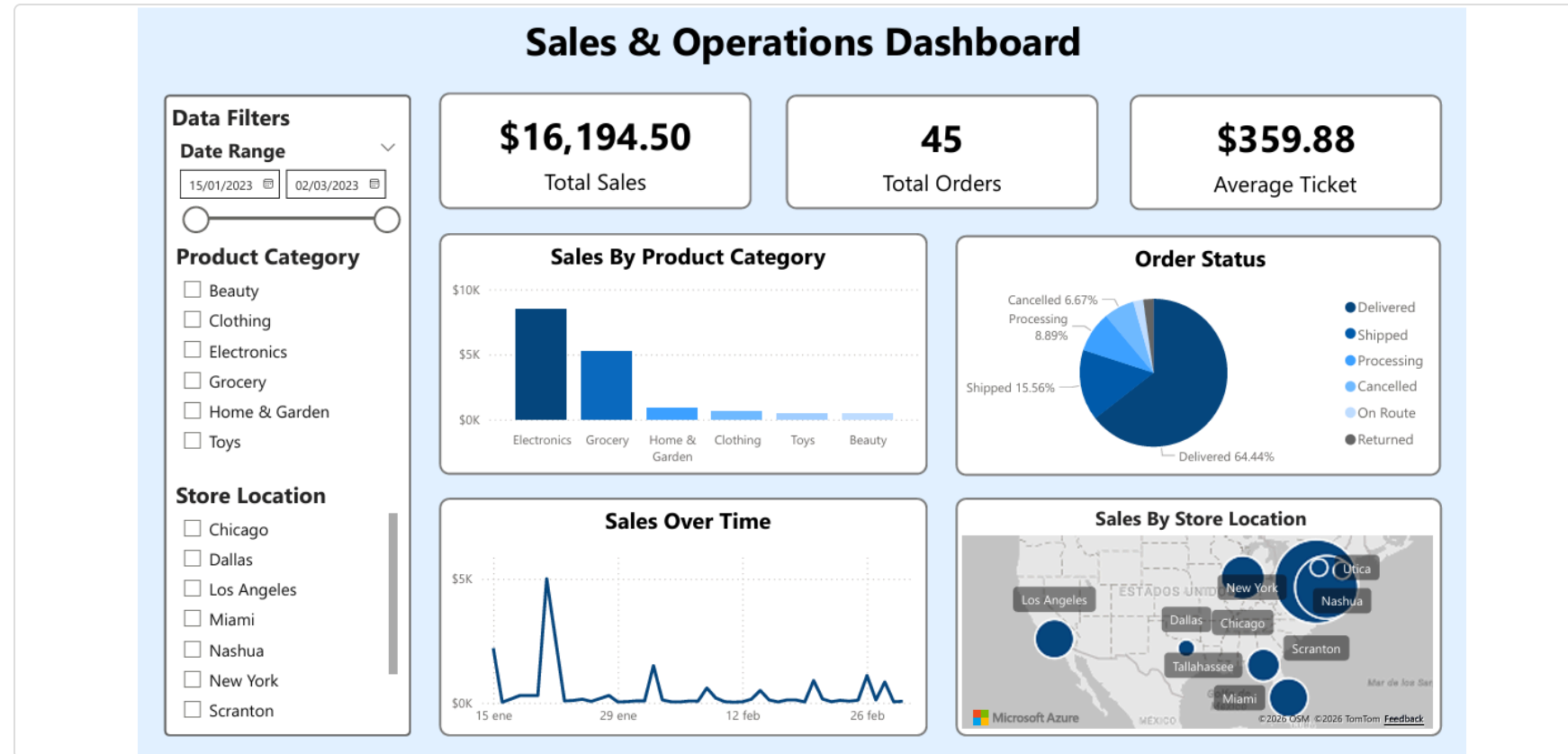
 Total Orders

45

 Average Ticket

\$359.88

Interactive Dashboard with Slicers



Product Categories

- **Electronics** leads (~\$9.6K)
- **Grocery** second (~\$5K)
- Minor: Home, Clothing

Order Status

- **Delivered:** 64.44%
- **Shipped:** 15.56%
- Proc: 8.9%, Cancel: 6.7%

Trends & Geography

- **Trends:** Peaks in Jan
- **Top:** NY, Miami, LA
- East/West coast concentration

Demonstrated Skills & Impact

Technical Competencies

**Data Cleaning**
Power Query, type normalization, catalog standardization, validations.

**Modeling & DAX**
Measure design for KPIs, slicers, relationships.

**Visual Analytics**
Creation of effective visualizations. Intuitive design.

**Automation**
Update from Excel, dynamic parameters.

**QA & Documentation**
Data dictionary and cross-validations.

Measurable Impact

Prep Time

87%

Efficiency

Consistency

>99%

Reliability

→ Next Steps

- Implement retention metrics (Cohorts)
- Publish to Power BI Service
- Integration with CRM/Inventory
- Develop critical KPI alerts